Project Proposal: Domain-Specific BPE Tokenizer for Medical Text

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Project Idea

I intend to implement a domain-specific Byte Pair Encoding (BPE) tokenizer trained on medical and clinical text, and evaluate how it impacts GPT model training and generation quality compared to the general-purpose byte-level BPE tokenizer from our assignments. Medical text presents a unique challenge for standard tokenizers: it is dense with specialized terminology (anatomical terms, drug names, abbreviations like "MRI," "CT," "DICOM"), compound words ("cholecystectomy," "intraoperative"), and shorthand conventions ("pt," "dx," "hx") that general-purpose tokenizers tend to fragment into suboptimal subword units. By training a BPE tokenizer directly on a medical corpus, I expect to learn a vocabulary that captures these domain-specific tokens as single units rather than splitting them across multiple tokens, which should lead to more compact tokenized sequences and potentially better downstream model performance on medical text.

How This Differs from the Assignment GPT

The GPT model from our assignments uses a byte-level BPE tokenizer (via HuggingFace's GPT-2 tokenizer retrained on our training data with a vocabulary size of 10,000). My variation replaces this tokenizer with one trained specifically on medical text while keeping the rest of the GPT architecture (attention, feedforward layers, positional embeddings, training loop) unchanged. This isolates the tokenizer as the single variable under study. The key difference is in what the model treats as its fundamental units of language: the domain-specific tokenizer should produce a vocabulary where common medical terms are represented as single tokens, while the general tokenizer would split them into multiple subwords. For example, a word like "laparoscopic" might be a single token in the medical vocabulary but split into "lap," "aro," "scop," "ic" by a general tokenizer. This means the model sees shorter sequences for the same text, effectively expanding its context window for medical documents. To ensure a fair comparison, I will control for vocabulary size between the two tokenizers and track both the number of tokens seen and the amount of text seen during training, since a more efficient tokenizer will pack more text into the same number of tokens per batch..

Experiments and Key Metrics

I plan to conduct the following comparisons between a GPT model using the domain-specific medical tokenizer and the baseline GPT with the general tokenizer. First, I will measure token efficiency: how many tokens each tokenizer produces for the same medical documents, and conversely, how they each perform on general English text (to see if specialization comes at a cost). Second, I will compare training convergence by tracking training loss over time for both models trained on the same medical corpus, to see whether the domain-specific tokenizer leads to faster or more stable convergence. Third, I will evaluate perplexity on held-out medical text as a quantitative measure of generation quality. Finally, I will perform a qualitative comparison of generated text from both models, prompting each with medical context and examining whether the outputs are coherent and use medical terminology appropriately. I am particularly interested in whether the vocabulary size tradeoff matters: a medical tokenizer may need a larger vocabulary to cover both general and medical terms, or it may sacrifice general-purpose coverage, and I want to understand how this tradeoff plays out. As an additional baseline, I will also evaluate the assignment's existing tokenizer retrained on the medical corpus (without vocabulary changes) to help distinguish whether any gains come from the domain-specific vocabulary itself or simply from training on in-domain data.

Datasets and Libraries

For the medical training corpus, I plan to use publicly available clinical text datasets such as the MTSamples collection (transcribed medical reports across various specialties) or PubMed abstracts, which are freely accessible and provide a large volume of domain-specific text. I will use Python with standard libraries (regex for preprocessing) and build the BPE implementation from scratch following the structure from our assignments, extending it to train on the medical corpus. PyTorch will be used for the GPT model training and evaluation, consistent with our prior assignments.